

# **Initiation à la recherche**

Méthodologie de la recherche

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# **Introduction to research**

Research methodology

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**Module «Projets, recherche et séminaires »**

**PR422 (Etu) : 2/4 / PR423 (App) : 2/4 ECTS**

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# Introduction of the speakers

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- **Vincent Berouille**, Course Coordinator
- Teacher/Researcher at Esisar and the LCIS Laboratory since 2002
- PhD in microelectronics from the University of Montpellier
- Director of LCIS
- Research activities focused on “Security and Safety of Integrated Circuits and Systems”

# Introduction of the speakers

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- **Oum-El-Kheir Aktouf**
- Teacher/Researcher at Esisar and the LCIS Laboratory since 1999
- PhD in computer science from the University of Grenoble
- Research activities focused on “Safety and security of autonomous and distributed systems”

# Preliminary information

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- Objectives of this chapter:
  - Understanding the scientific method
    - for Embedded Systems, and Information and Communication Technologies (ICT)
    - **experimental research**
  - Writing a scientific paper

# Plan

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- Introduction
  - Critical thinking
  - The main steps of the method
  - Reasoning approaches
- 
- Writing a scientific paper

# Plan

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- Introduction
  - What is the scientific method ?
  - Research question
  - State of the Art
  - Peer review

# What is the Scientific Method?

## (in experimental research)

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- A **systematic process** to build reliable knowledge
- Based on observation, experimentation, and reasoning

# What is the Scientific Method?

## Key Features

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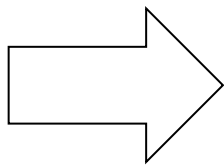
- **Evidence-based** and **testable**
- **Repeatable** by others
- Open to revision
- Communicable to the scientific community

# What is the Scientific Method?

## Limits and Challenges

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- Human and technical errors
- Scientific integrity and context factors (pressure, conflicts, ethics)
- **Bias**
- Complexity of natural phenomena
- Knowledge evolves over time



**Critical thinking**

**Skepticism**

# Research Question

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- Clear / Precise / Concise / Easy to understand
- Raises from a critical analysis of the state-of-the-art
- Addresses :
  - A lack of knowledge
  - A contradiction
  - A methodological difficulty

**Defines the purpose and direction  
of your study**

# State of the Art

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- **A good literature review helps to:**
  - **Avoid wasting time** on problems already solved
  - Identify what has already been done
  - Spot **open questions** and research gaps
  - Avoid repeating the same mistakes
  - Position your work in the existing context

**Saves time and sharpens  
your research focus.**

# Peer Review

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- Helps **detect errors, flaws, or biases**
- Ensures **scientific quality and credibility**
- Provides **independent and constructive feedback**
- Improves the clarity, rigor, and reliability of the work

**Publication and peer review strengthen research integrity and quality.**

# Plan

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- Critical thinking
  - Biases
  - Ensuring research accuracy

# Critical Thinking

## *Key biases in scientific research papers*

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- Main (Unconscious) Biases:
  - **Confirmation bias**  
Focusing on results that support the expected hypothesis.
  - **Selection bias**  
Selecting data, samples, or studies that favor a specific outcome.

# Critical Thinking

## *Key biases in scientific research papers*

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- **Publication bias**

Positive results are more likely to be published.

- **Authority bias**

Accepting claims without question because they come from recognized experts.

- **Overinterpretation bias**

Overstating the meaning or impact of the results.  
(e.g., concluding that a correlation between two variables proves a causal relationship...)

**And many other biases !**  
**Common sense is not enough !**

# Critical Thinking

## Ensuring Research Accuracy

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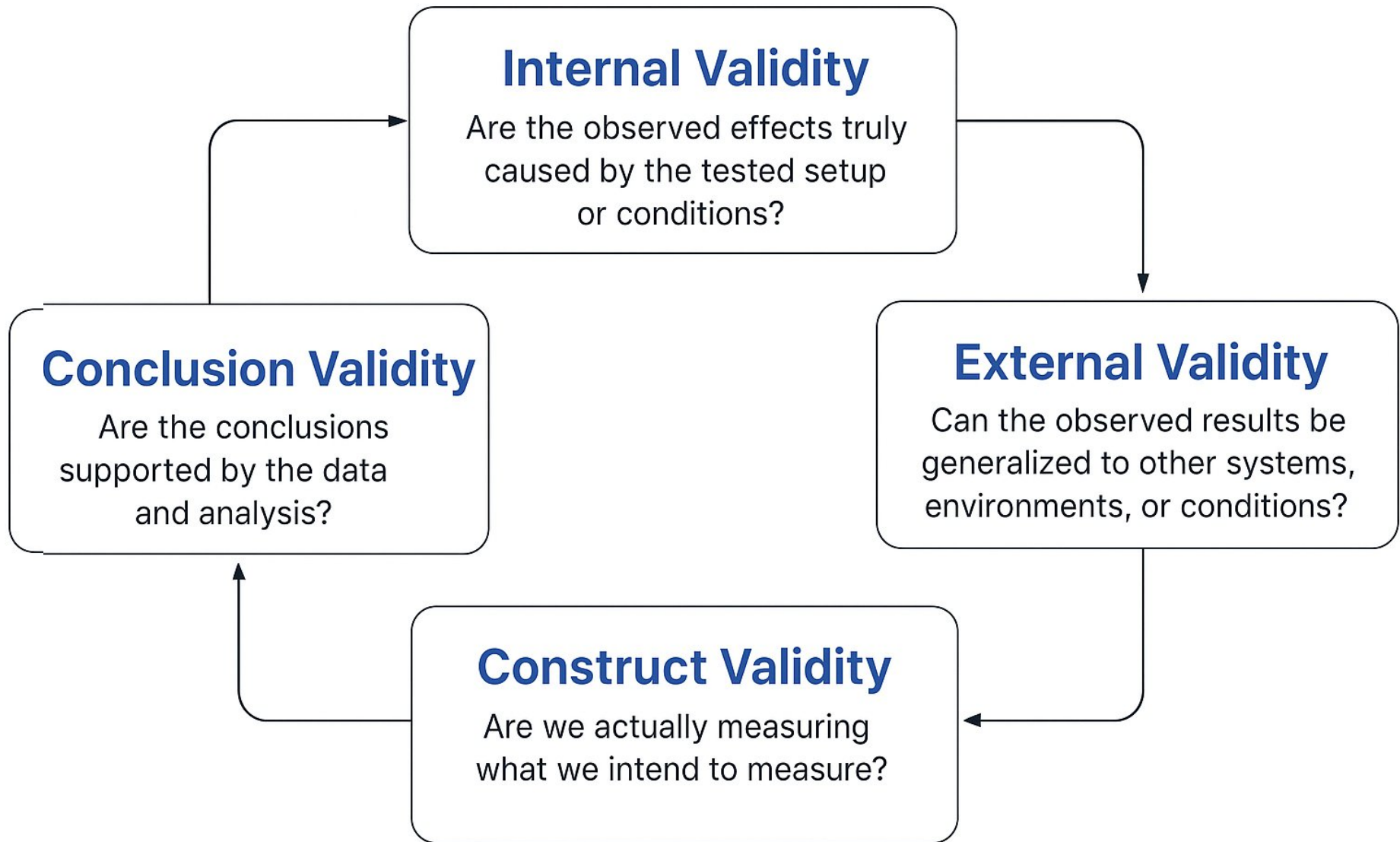
- *Internal validity*
  - Are the observed effects truly caused by the tested setup or conditions?
- *External validity*
  - How applicable are the results beyond the tested setup?

# Critical Thinking

## Ensuring Research Accuracy

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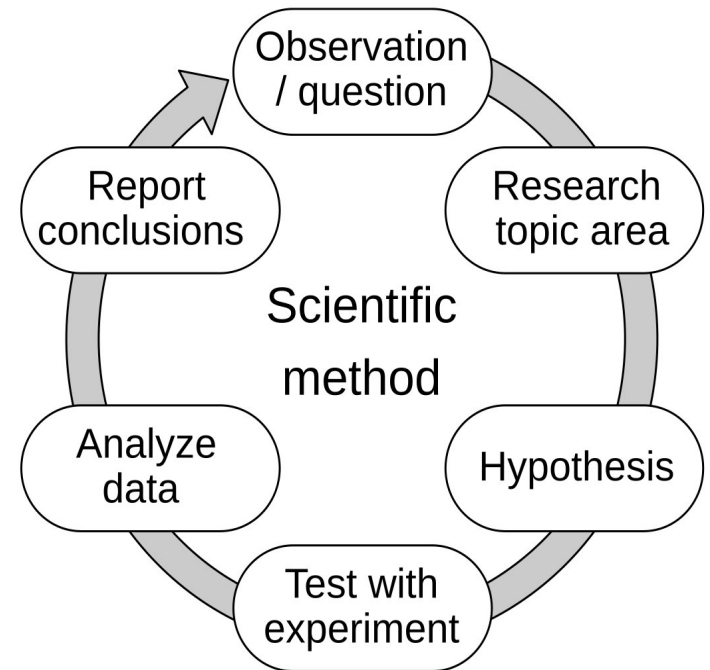
- *Construct validity*  
Do the metrics and tests accurately represent the concepts under study? (e.g., sensor accuracy vs bandwidth)
- *Conclusion validity*  
Are our conclusions reliably supported by the data and analysis?



# The Main Steps of the Method (simplified version)

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- Observation
- Question
- Hypothesis (falsifiable)
- Predictions
- Experiment / Test
- Analysis & Conclusion



Must be repeated and refined as needed:

**New experiments increase or decrease the level of confidence in a theory**

# Experiment / Test

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- Key principles:
  - Must be **reproducible** by others
  - All steps, parameters, tools, and conditions must be **clearly described**
  - Allows other researchers to **verify, test, or challenge** your results
  - The goal of a test is to try to **refute the theory**, not to verify it (e.g., increase the workload to test the real time task scheduler)

**Transparency = Scientific credibility**

# Reasoning Approaches

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- Inductive reasoning

**From specific observations  
to general conclusions**

Observation: Circuits fail under a specific electromagnetic disturbance.

General conclusion: This type of disturbance causes errors in these circuits.

# Reasoning Approaches

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- Deductive Reasoning

**From general principles to specific predictions**

General principle: Ohm's law  $V=RxI$

Specific prediction: For a resistor of  $10\ \Omega$ , applying  $5\ V$  will produce a current of  $0.5\ A$ .

# Reasoning Approaches

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- Abductive Reasoning

**From incomplete observations to the most likely explanation**

Incomplete observation: A circuit intermittently stops working, but only under high temperature.

Most likely explanation: The components overheat, causing the malfunction

# Mathematical Modeling

## Key Principles

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- Mathematics: based on axioms and logic, no observation is needed to validate them.
- Experimental sciences: observe the real world to test hypotheses and theories.

**But Mathematics is used to model physical phenomena, and these models are then validated through experiments.**

# Mathematical Modeling

## Key Principles

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- Often starts with an initial hypothesis. Then, build and validate a **simplified abstract model** of the real system.
- Once validated, the model is used to test ideas through **calculations or simulations** instead of physical experiments.

**Models can predict behavior without  
any actual implementation**

# Mathematical Modeling

## Key Principles

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- **Two essential points:**
  - **Simplification / Abstraction:**  
Keep only what matters in the system.
  - **Correspondence Rules:**  
Define how the model's results should be interpreted to describe the real world.

**A good model balances simplicity and relevance.**

# Mathematical Modeling

## Key Principles

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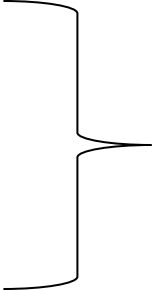
- **Types of reasoning used:**
  - **Deductive (most common):**  
From the model deduce results.
  - **Inductive:**  
Generate data (e.g. via Monte Carlo simulations) to identify trends or patterns.
  - **Abductive:**  
Propose the model that best explains the available observations.

# Writing a scientific paper : the plan

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- Titre / Authors / Affiliations
- Abstract
- Key words

1. Introduction
2. Previous work
3. Methodology
4. Experimental setup
5. Result discussion
6. Conclusion and perspectives
7. Acknowledgement
8. References



The discussion of results is often overextended compared to the time actually spent on the research work

# Writing a scientific paper

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- The **title** must represent the paper's content: concise, accurate, and not misleading to the reader.
- **Authors** are listed by **order of importance** for the contribution (the first author is the main contributor, often the PhD student)
- **Abstract**: a short summary without technical details, easy to read for non-specialists.
- **Keywords**: the most important words, fewer than 10.

# Writing a scientific paper

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- Introduction
  - **The most important chapter**
  - Context description (not too general)
  - Motivations
  - Research question description
  - Contributions of the paper
  - Description of the paper organization (In the following)

# Writing a scientific paper

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- Methodology
  - A chapter dedicated to theoretical considerations
  - No technical details
  - Explanation of the methods to obtain results, and the verification process
  - Procedure or test plan description

**The goal of this chapter is to focus on  
theoretical aspects  
without interfering with practical details**

# Writing a scientific paper

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- Experimental setup
  - Accurate presentation of the experimental setup : with all details to assure reproducibility
  - Result presentation

**The goal of this chapter is to give all information necessary to reproduce exactly the same experiment**

# Writing a scientific paper

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- Result discussion
  - Interpreting main results
  - Result importance
  - Comparing theoretical results and experimental results
  - Discussing the biases and the threats to the validity
  - Discussing limitations

# Writing a scientific paper

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- Conclusion and perspectives
  - **The second most important chapter**
  - Summarize all the important results
  - Give 2 or 3 short-term or mid-term perspectives

# Writing a scientific paper

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- All figures must be self-explanatory with their captions (some readers only read figures first...)
- However, always describe the figure in the text to help the reader understand them without effort (In Fig. 1, ...).

# Writing a scientific paper

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- English should be **precise and clear, like mathematics.**
- Always **define new terms** when introduced.
- Use **conjunction words** to explain your reasoning (e.g., *as, thus, indeed, although*).
- Justify your assertions (with references when possible)
- Use the **exact same words consistently** rather than replacing them with synonyms.

# Writing a scientific paper

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Write it

Wait to forget it

Read it

Rewrite it

Wait to forget it

Rewrite it

...

Repeat until « no more time »...